

The title of my awesome paper

Author1¹, Author2², Author3², and Matías Mattamala³

Abstract—(Matias) One brief sentence giving context

(Matias) One sentence on the problem and why it is worth solving

(Matias) One sentence introducing this work. Other sentence with main contributions

(Matias) Two sentences with main results, highlighting some key numerical result or achievement

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I. INTRODUCTION

(Matias) First paragraph should give the context

Robots are becoming a promising assistive platform in different societal context...

(Matias) Second paragraph should talk about the problem that we are concerned about, given the context

However, implementing assistive robotic solutions present different challenges because of human-robot interaction...

(Matias) Third paragraph raises the research gap that exist to solve the problem

Previous methods have attempted to solve this problem but they fall short at certain things...

(Matias) Fourth paragraph presents our solution to address the gap

In this paper, we present an assistive autonomy system that addresses these challenges by...

(Matias) Last paragraph indicates the contributions our solution *The contributions of this work are:*

- *Contribution 1* (this needs to be demonstrated in an experiment)
- *Contribution 2* (this needs to be demonstrated in an experiment)
- The source code is available at <https://www.github.com>

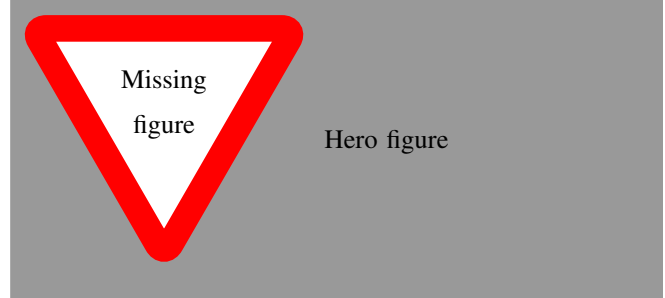


Fig. 1. **Hero figure.** This figure shows an illustration of the approach.

II. RELATED WORK

(Matias) Here we want to cover related works in at least two areas of interest. We can use subsections to split it

(Matias) A good related work does not merely list papers (that's a bad related work section), but instead comprehensively explains the most relevant works that overlap with the topics of this paper.

(Matias) For each paper, we want to (1) explain what they did that it was relevant, and (2) suggest some shortcoming that we address in this paper

(Matias) We need to close each subsection explaining how this paper addresses the main shortcomings of prior work. In that way, we make sure that the readers understand the gap we are filling.

(Matias) Lastly, as a personal advice, I recommend against citing papers using the reference number as a noun. For example, instead of saying “[1] introduced a comprehensive review on interactive perception”, we say “Bohg et al. [1] introduced a comprehensive review on interactive perception”

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III. METHOD

(Matias) Here we introduce the main technical novelties

(Matias) I strongly recommend adding a block diagram, because robotics papers are usually about systems. The block diagram should indicate the main components, without providing excessive details

(Matias) Then, the subsections of this section should correspond to blocks in the diagram. In that way it is very easy for the readers to always come back to the block diagram to get the big picture, and grasp the details in the corresponding subsection

(Matias) Additionally, the subsections can have additional figures to illustrate specific steps or sub-pipelines required

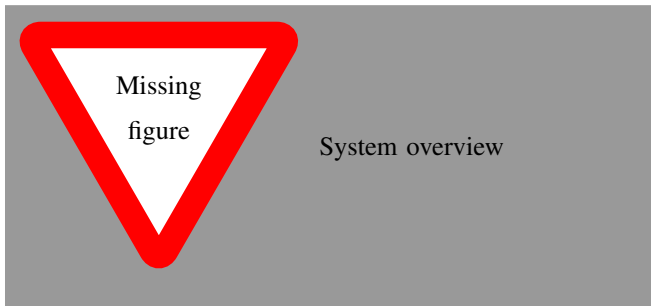


Fig. 2. **System overview.** This shows the block diagram of the solution.

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A. Module 1

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B. Module 2

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C. Module 3

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IV. RESULTS

(Matias) We should first provide an overview on the experiments we did to assess the contributions

A. Experimental Setup

(Matias) I recommend having this section where we explain some stuff required to understand the experiments, for example:

Robot platform: (Matias) If we did experiments with a specific robot(s)

Evaluation platform: (Matias) If the experiments were done in simulation or with datasets, we need to specify the computer specs where the software was run on

Datasets: (Matias) We specify all the datasets, either existing, or manually collected. If manually collected, indicate the protocol. We indicate the naming convention for the sequences. We indicate the splits (if relevant). Indicate how the ground truth was obtained and accuracy (if pertinent)

Evaluation metrics: (Matias) We introduce all the relevant metrics and analytical expressions if needed (e.g., if we are introducing a new one)

Baselines: (Matias) We introduce all baselines we used. They can be introduced here or in the relevant experiments

B. Experiment I – Simulation

(Matias) Each experiment needs to explain the purpose and what we aim to prove with it. This should respond to one (or multiple) claimed contributions

(Matias) Here we explain details of which specific datasets, metrics, baselines

(Matias) We provide a description on the findings and main conclusions of the experiment

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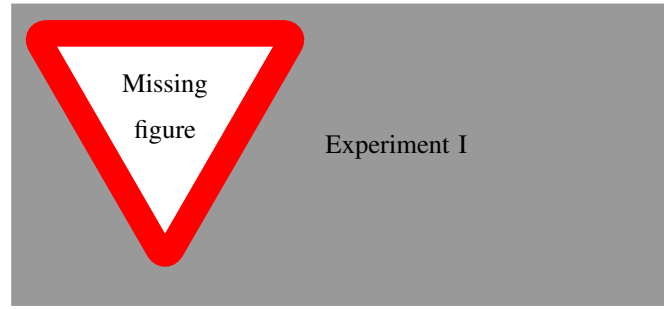


Fig. 3. **Experiment I – Simulation.** We show the main results obtained in this experiment. The results show that our method achieves better performance...

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C. Experiment II – Real-world

(Matias) Just another placeholder for another experiment

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D. Experiment III – Demonstration

(Matias) Sometimes we also want to describe a qualitative experiment demonstrating a downstream capability

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V. CONCLUSION

(Matias) Conclusion summarises the main findings of the work, discussion on limitations, and possibly future work

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- [1] J. Bohg, K. Hausman, B. Sankaran, O. Brock, D. Kragic, S. Schaal, and G. S. Sukhatme, "Interactive perception: Leveraging action in perception and perception in action," *IEEE Trans. Robot.*, vol. 33, no. 6, pp. 1273–1291, 2017. [Online]. Available: <https://doi.org/10.1109/TRO.2017.2721939>